NATIONAL INSTITUTE OF TECHNICAL TEACHERS' TRAINING & RESEARCH, KOLKATA

Deemed to be University

Department of Computer Science & Engineering

Program: M. Tech. in AI & ML with Pedagogy

DOCUMENTATION OF

ASSIGNMENT - 3

Course Name: Artificial Intelligence Lab

Course Code: AIMLPPC107

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Semester: 1st

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Session: 2026-27

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Faculty's Signature **Student's Signature**

1. PROBLEM STATEMENT

After a flood, an emergency-control centre must plan rescue operations. Different routes and rescue facilities are available, but some activities such as evacuation, medical support and transportation may all be required. Develop a system to generate a feasible and cost-effective response plan.

2. BRAINSTORMING

The brainstorming process focused on developing a systematic approach to planning emergency response operations after a flood. The problem was analysed by identifying affected locations, available facilities, emergency tasks, alternative rescue methods and task dependencies.

An AND/OR graph was selected because it can represent both mandatory activities and alternative courses of action. A Python backend using FastAPI was chosen to process the planning logic, while HTML, CSS and JavaScript provide an interactive web interface.

The system accepts a budget and planning priority, generates a candidate response plan, checks task dependencies, calculates estimated costs and displays the result.

2.1 LIST OF ACTIVITIES

The following activities can be performed to solve the problem:

- Identify the affected locations and available emergency facilities.
- Define the emergency tasks required after a flood.
- Assign estimated costs and beneficiary counts to each task.
- Represent task dependencies using an AND/OR graph.
- Identify mandatory tasks and alternative response options.
- Configure the available response budget.
- Generate a candidate response plan.
- Check whether the prerequisites of selected tasks are satisfied.
- Calculate the total estimated cost and beneficiary count.
- Display the AND/OR graph and task details.
- Compare the estimated cost against the available budget.
- Export the generated response plan as a CSV file.

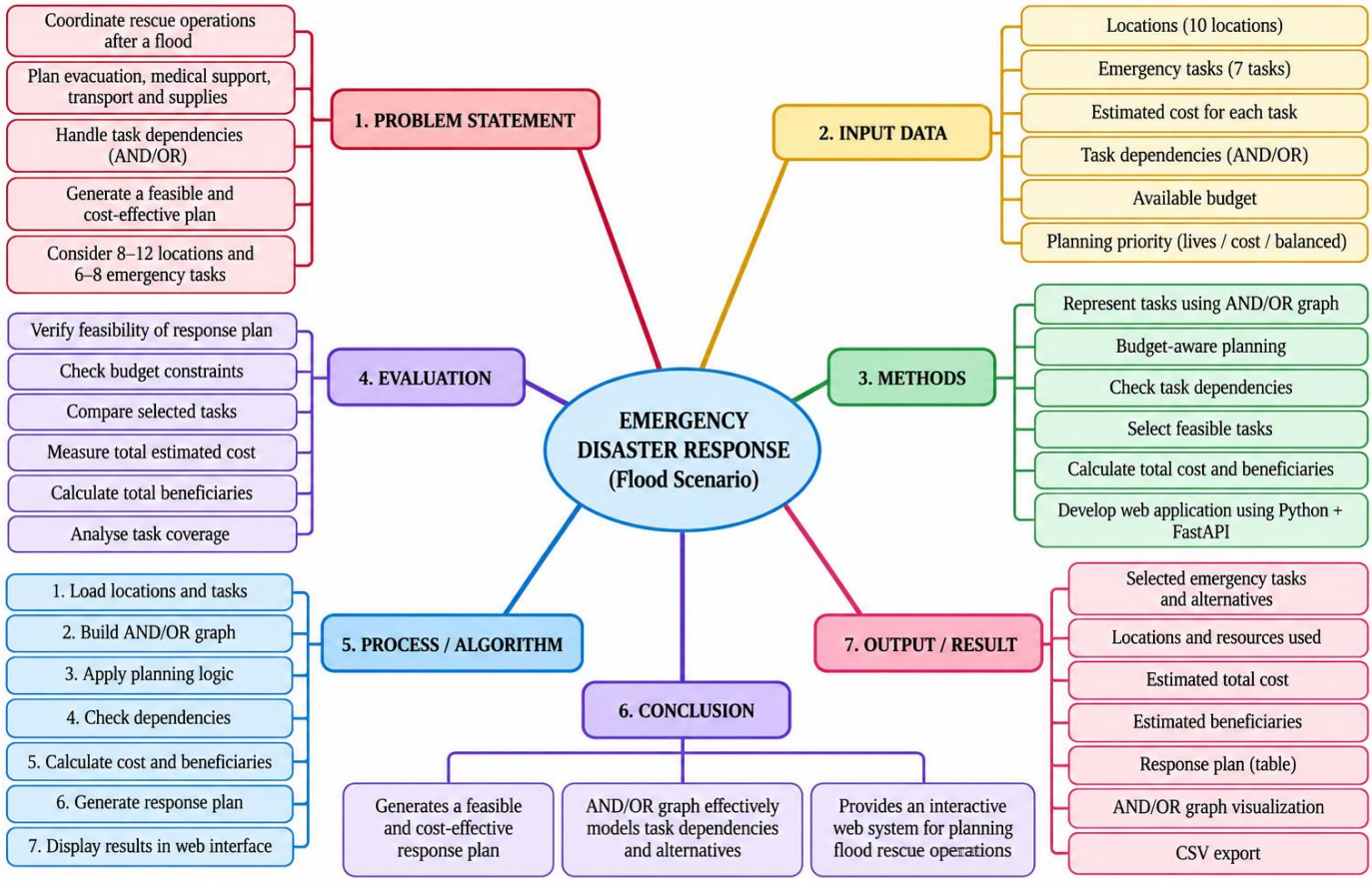
2.2 OBJECTIVES

The objectives need to achieved at the end of solving this problem:

- To understand the working mechanism of AND/OR graphs.
- To model flood emergency response as a graph-based planning problem.
- To represent mandatory task dependencies and alternative actions.
- To develop a web application using Python and FastAPI.
- To generate a candidate plan under a configurable budget.
- To display task coverage, estimated costs and beneficiary counts.
- To provide an interactive graph and node inspection facility.
- To export the generated plan for further analysis.

2.3 CONCEPT MAP

The Emergency Disaster Response system begins with the overall objective of coordinating a flood rescue operation. The objective is decomposed into smaller emergency tasks using an AND/OR graph. The main task groups include evacuation, medical support, patient transportation, clean-water distribution, shelter activation and area assessment.



3. MATHEMATICAL MODELING

The emergency response problem is represented using a directed AND/OR graph. Each node represents an objective or emergency task, and the edges represent task decomposition or dependency relationships.

The overall objective is to develop a feasible response plan that covers essential emergency activities while keeping the estimated cost within the available budget.

3.1 REPRESENTATION OF LOCATIONS

Let the set of locations be:

$$V = \{L_1, L_2, \dots \dots, L_{10}\}$$

The locations are represented as follows.

Location ID	Location name	Type
L1	Control Centre	Command and assessment
L2	North Shelter	Temporary shelter
L3	Riverside Clinic	Medical facility
L4	Central Depot	Supply depot
L5	East Bridge	Route/crossing

Location ID	Location name	Type
L6	Lowland Village	Flood-affected area
L7	Market District	Flood-affected area
L8	South School	Temporary shelter
L9	West Landing	Boat access
L10	Water Plant	Water supply

These locations are illustrative and represent the facilities involved in a hypothetical flood-response scenario.

3.2 REPRESENTATION OF EMERGENCY TASKS

Let the set of emergency tasks be:

$$\boldsymbol{T} = \{\boldsymbol{T_1}, \boldsymbol{T_2}, \dots\dots, \boldsymbol{T_T}\}$$

Each task has an estimated cost, an estimated number of beneficiaries, a location, a logical type and a set of prerequisites.

Task ID	Emergency task	Type	Cost (₹)	Estimated beneficiaries
T1	Assess affected zones	AND	900	0
T2	Evacuate Lowland Village	OR	4,200	310
T3	Evacuate Market District	OR	2,600	180
T4	Set up medical support	AND	1,800	70
T5	Transport critical patients	OR	2,200	24
T6	Distribute clean water	AND	1,500	500
T7	Open temporary shelters	OR	1,100	670

The costs and beneficiary estimates are hypothetical values used for academic demonstration.

3.3 AND/OR GRAPH REPRESENTATION

An AND/OR graph is a representation in which a problem can be decomposed into smaller subproblems. It is useful when some objectives require several tasks to be completed together, while others can be achieved through alternative actions.

AND node

An AND node indicates that all required child tasks or prerequisites must be satisfied.

For example, medical support may require both a medical team and the necessary patient-transfer arrangements. Similarly, water distribution may depend on water-delivery arrangements and appropriate distribution facilities.

If the required subtasks are $\boldsymbol{A}$ and $\boldsymbol{B}$, then:

$$AND(A, B) = A \wedge B$$

Both A and B must be satisfied.

OR node

An OR node indicates that one feasible alternative can be selected.

For example, the evacuation of Lowland Village may use either:

- Boat evacuation through West Landing.
- A bus route through East Bridge, if the route is safe.

The logical representation is:

$$OR(A, B) = A \vee B$$

At least one valid alternative must be selected.

3.4 TAKE DEPENDENCY MODEL

The task dependencies in the proposed system are:

Task	Prerequisite
T1 — Assess affected zones	None
T2 — Evacuate Lowland Village	T1
T3 — Evacuate Market District	T1
T4 — Set up medical support	T1
T5 — Transport critical patients	T4
T6 — Distribute clean water	T1 and T7
T7 — Open temporary shelters	T1

These dependencies express the simplified planning assumptions used by the application.

For example, patient transportation depends on medical support, while clean-water distribution depends on both assessment and shelter activation.

3.5 COST MODEL

Let c_i represent the estimated direct cost of task T_i.

If S is the set of selected tasks, the total estimated cost is:

$$C(S) = \sum_{T_i \in S} c_i$$

Let B be the available budget.

The budget constraint is:

$$C(S) \leq B$$

For example, if the available budget is ₹18,000 and all seven tasks are selected, the total estimated cost is::

$$C(S) = 900 + 4200 + 2600 + 1800 + 2200 + 1500 + 1100$$
$$C(S) = 14300$$

Since ₹14,300 is less than ₹18,000, the selected tasks satisfy the budget constraint. However, satisfying the budget alone does not guarantee that the plan is optimal. The system must also evaluate alternative strategies and select a feasible, minimum-cost solution.

3.6 AO* COST EVALUATION MODEL

The AO* algorithm evaluates the AND/OR graph to identify a minimum-cost solution graph. It expands selected nodes, evaluates their child nodes, and updates the estimated costs towards the root node.

Let f(n) represent the estimated cost of solving node n, c(n) represent its own cost, and S(n) represent its child nodes.

For an AND node, all required child nodes must be included in the solution. Its cost is calculated as:

$$f(n) = c(n) + \sum_{s \in S(n)} f(s)$$

For an OR node, the lowest-cost feasible alternative is selected. Its cost is calculated as:

$$f(n) = c(n) + \min_{s \in S(n)} f(s)$$

Example:

Suppose Lowland Village can be evacuated by either boat or bus. The estimated cost of boat evacuation is ₹4,500, while the estimated cost of bus evacuation is ₹3,200. Assuming both alternatives are safe and have sufficient capacity, the OR-node cost is:

$$f(T_2) = min(4,500, 3,200)$$
$$f(T_2) = ₹3,200$$

Therefore, the bus is selected because it is the cheaper feasible alternative.

For a medical response requiring both medical support and patient transportation, the AND-node cost is:

$$f(T_4) = c(T_4) + f(T_5)$$
$$f(T_4) = 1,800 + 2,200$$
$$f(T_4) = ₹4,000$$

This calculation assumes that medical support and patient transportation are both required and that their costs are counted separately.

3.7 AO* SEARCH PROCEDURE

The AO* search procedure explores the AND/OR graph to determine the preferred solution graph. The procedure begins with the root node representing the overall flood-response objective. Initial cost estimates are assigned to unexpanded nodes, and the most promising partial solution graph is selected for expansion.

The selected node is expanded to generate its child nodes, representing required subtasks or alternative strategies. The costs are evaluated using the AND cost-summation rule or the OR minimum-cost rule. The

revised costs are propagated towards the root, and the preferred solution graph is updated whenever a better alternative is found.

This process continues until the best solution graph is established under the algorithm's assumptions. The final plan is then checked against the task dependencies, operational constraints, and available budget. For an optimality guarantee, the implementation must use appropriate heuristic estimates and correctly perform node expansion and cost updates.

3.8 TASK DEPENDENCY AND FEASIBILITY CONSTRAINTS

The response plan must satisfy the task dependencies defined in Section 3.4. For example, patient transportation cannot be treated as completed before the required medical-support task is completed. Similarly, clean-water distribution depends on both assessment and shelter activation.

Let $\boldsymbol{x_i}$ be a binary variable indicating whether task $\boldsymbol{T_i}$ is selected. If $\boldsymbol{x_i = 1}$, the task is selected; if $\boldsymbol{x_i = 0}$, the task is not selected.

For the dependency $\boldsymbol{T_1 \rightarrow T_2}$, the constraint is:

$$\boldsymbol{x_2 \leq x_1}$$

For the dependency $\boldsymbol{T_4 \rightarrow T_5}$, the constraint is:

$$\boldsymbol{x_5 \leq x_4}$$

For clean-water distribution, which depends on both assessment and shelter activation, the constraints are:

$$\boldsymbol{x_6 \leq x_1}$$

$$\boldsymbol{x_6 \leq x_7}$$

These constraints ensure that a dependent task is selected only when its prerequisites are also selected.

Operational feasibility must also be considered. For example, a bus route can be selected only if it is safe and has sufficient capacity to transport the required number of people. The system should reject infeasible strategies even if they have a lower estimated cost.

3.9 WORKED NUMERICAL EXAMPLE

Consider a flood-response scenario with an available budget of $₹\mathbf{10,000}$. The response requires assessment of the affected area, evacuation of $\mathbf{80}$ people, medical support, and transportation of critical patients.

The estimated cost of assessment by a survey team is $₹\mathbf{500}$. Boat evacuation costs $₹\mathbf{4,500}$, while bus evacuation costs $₹\mathbf{3,200}$. Medical support costs $₹\mathbf{1,000}$, and ambulance transportation costs $₹\mathbf{1,500}$.

Assume both evacuation methods are safe and can accommodate all 80 people.

Step 1: Evaluate the OR node.

The system compares boat and bus evacuation and selects the lower-cost feasible alternative.

$$\boldsymbol{C(evacuation) = min(4,500, 3,200)}$$

$$\boldsymbol{C(evacuation) = ₹3,200}$$

Therefore, the bus is selected.

Step 2: Evaluate the required medical activities.

Medical support and patient transportation are both required. Their combined cost is:

$$\boldsymbol{C(medical\ response)\ =\ 1,000\ +\ 1,500}$$

$$\boldsymbol{C(medical\ response)\ =\ ₹2,500}$$

Step 3: Calculate the total response cost.

The complete response requires assessment, evacuation, medical support, and patient transportation.

$$\boldsymbol{C(total)\ =\ 500\ +\ 3,200\ +\ 1,000\ +\ 1,500}$$

$$\boldsymbol{C(total)\ =\ ₹6,200}$$

Step 4: Verify the budget.

The available budget is $₹\mathbf{10,000}$, and the estimated response cost is $₹\mathbf{6,200}$. Therefore:

$$\boldsymbol{₹6,200\ \leq\ ₹10,000}$$

The budget constraint is satisfied. The remaining budget is:

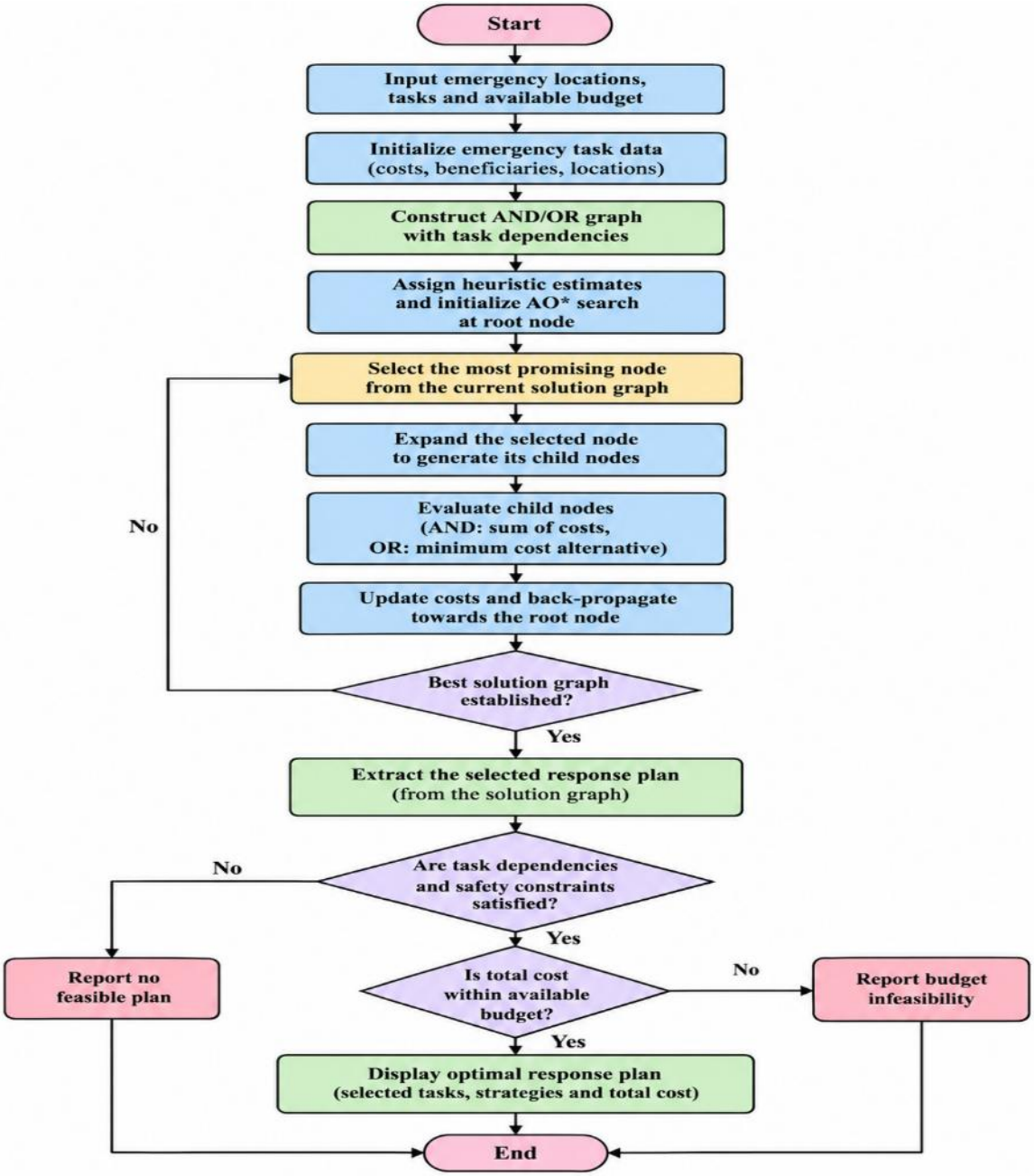
$$\boldsymbol{Remaining\ budget\ =\ 10,000\ -\ 6,200}$$

$$\boldsymbol{Remaining\ budget\ =\ ₹3,800}$$

Thus, the selected plan consists of the survey team, bus evacuation, medical support, and ambulance transportation, with an estimated total cost of $₹\mathbf{6,200}$.

This example illustrates AND/OR cost evaluation and budget verification. A complete AO* implementation must additionally perform graph expansion, heuristic evaluation, and cost back-propagation to establish the solution graph rather than simply selecting the cheapest option.

4. FLOWCHART



5. IMPLEMENTATION

The system consists of three main components: the user interface, the backend planning module, and the emergency task database.

The user interface allows users to view emergency locations, task details, costs, dependencies, and the available budget. The backend receives the planning request and processes the task data. The planning module evaluates the AND/OR graph and calculates the estimated cost of the selected activities. Finally, the application displays the resulting response plan and its budget status.

5.1 WORKING OF SYSTEMS

The system begins by loading the predefined emergency locations and task information. Each task contains an identifier, description, estimated cost, beneficiary estimate, and prerequisite relationships.

The system then constructs the AND/OR graph according to the task structure. AND nodes represent activities that require multiple subtasks, while OR nodes represent activities with alternative strategies. The planning algorithm evaluates the graph and updates the estimated costs of its nodes.

After evaluating the available strategies, the system identifies a candidate response plan. It checks the task dependencies and total estimated cost against the available budget. The final output includes the selected tasks, estimated cost, and budget status.

5.2 ALGORITHM

The planning procedure follows these steps:

Step 1: Start the application and load the emergency task data.

Step 2: Receive the available budget and planning requirements.

Step 3: Construct the AND/OR graph using the task relationships and dependencies.

Step 4: Initialize the search from the root node representing the overall emergency-response objective.

Step 5: Evaluate AND nodes by combining the costs of all required child nodes.

Step 6: Evaluate OR nodes by comparing alternative strategies and selecting the lowest-cost feasible alternative.

Step 7: Update node costs and propagate the revised estimates towards the root.

Step 8: Continue the search until the solution graph is established.

Step 9: Validate the selected plan against the dependencies, operational constraints, and available budget.

Step 10: Display the selected response plan and its estimated total cost.

5.3 TEST CASES

Test Case 1: Budget sufficient for all required tasks.

Input: Available budget of ₹18,000.

Expected result: The complete task set can satisfy the budget constraint if all required activities are feasible.

Test Case 2: Selection between evacuation alternatives.

Input: Boat evacuation costs ₹4,500, while bus evacuation costs ₹3,200.

Expected result: The bus is selected if both alternatives are safe and have sufficient capacity.

Test Case 3: Task dependency validation.

Input: Patient transportation is requested without completing the required medical-support task.

Expected result: The system identifies the dependency violation and does not accept the incomplete plan.

Test Case 4: Budget insufficient for the selected plan.

Input: The estimated cost of the complete selected plan exceeds the available budget.

Expected result: The system reports budget infeasibility or searches for another valid alternative. Mandatory emergency activities should not be silently removed.

Test Case 5: Alternative strategy becomes infeasible.

Input: The bus route is unsafe or lacks sufficient capacity.

Expected result: The bus strategy is rejected, and the planner evaluates other feasible alternatives.

5.4 CODING

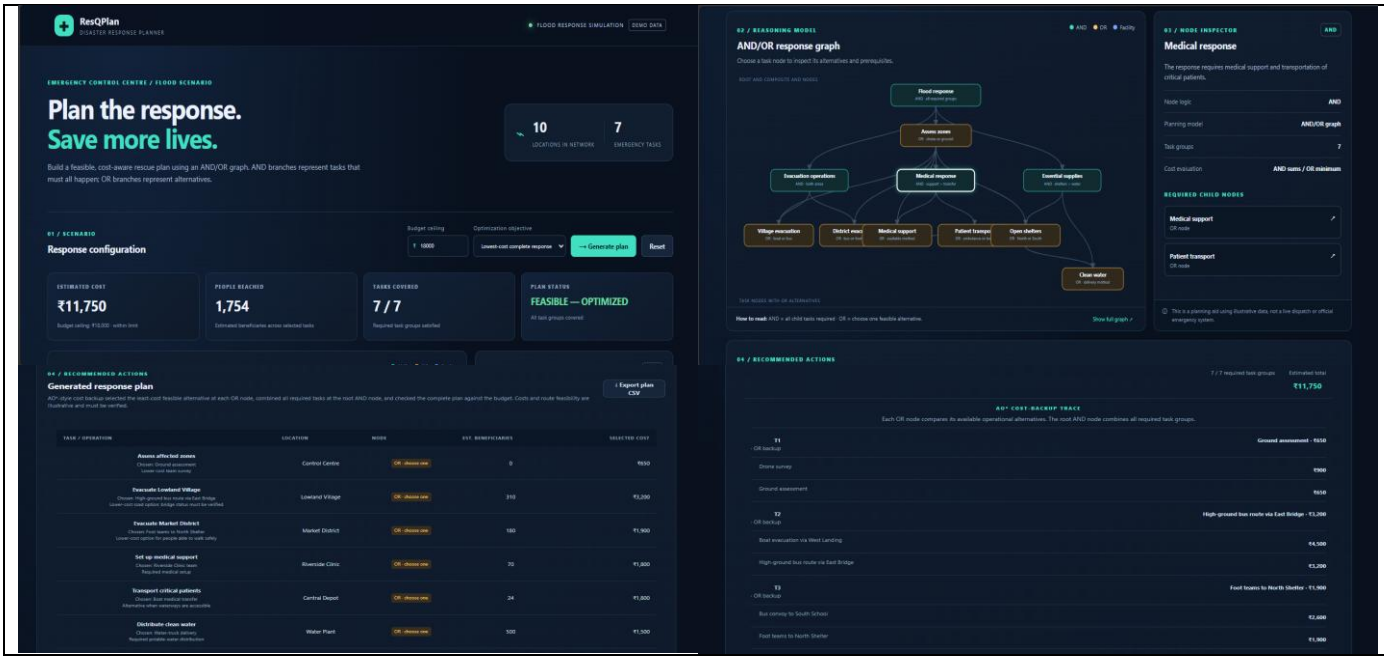
The ResQPlan application is implemented using Python and FastAPI. The program separates emergency task data, planning logic, and web interface functionality into different modules. The data.py file stores the locations and emergency tasks, planner.py evaluates the available strategies, and app.py provides the API endpoints used by the frontend.

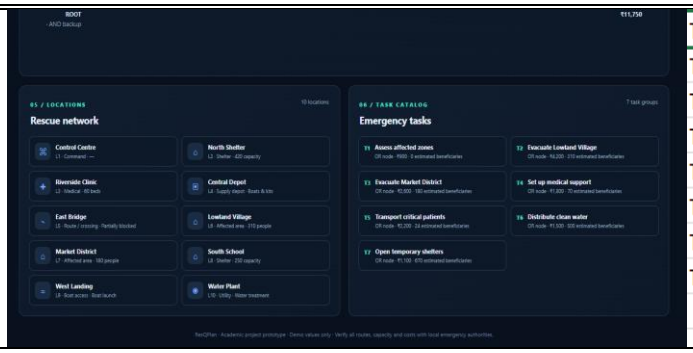
```
def _select_option(task, priority):
    """Select the least-cost feasible alternative."""
    options = [
        option
        for option in OPTIONS[task["id"]]
        if option["feasible"]
    ]
    if not options:
        return None
    return min(
        options,
        key=lambda option: (option["cost"], option["name"])
    )
def generate_plan(
    budget: int = 18000,
    priority: str = "balanced"
) -> dict:
    tasks_by_id = {
        task["id"]: task for task in TASKS
    }
    selected = {}
    trace = []
    # Evaluate alternatives at each OR node
    for task in TASKS:
        option = _select_option(task, priority)
        if option is None:
            continue
        selected[task["id"]] = option
        trace.append({
            "node": task["id"],
            "operation": "OR backup",
            "chosen": option["name"],
            "backed_up_cost": option["cost"]
        })
    # Calculate the total cost at the root AND node
    tasks_result = []
    for task in TASKS:
        option = selected.get(task["id"])
        if option is None:
            continue
        item = dict(task)
```

```
        item["option"] = option["name"]
        item["cost"] = option["cost"]
        tasks_result.append(item)
    total_cost = sum(
        task["cost"] for task in tasks_result
    )
    all_covered = (
        len(tasks_result) == len(TASKS)
    )
    within_budget = total_cost <= budget
    feasible = all_covered and within_budget
    trace.append({
        "node": "ROOT",
        "operation": "AND backup",
        "combined_cost": total_cost,
        "budget": budget,
        "result": (
            "complete and within budget"
            if feasible
            else "budget exceeded"
        )
    })
    return {
        "tasks": tasks_result,
        "total_cost": total_cost,
        "budget": budget,
        "feasible": feasible,
        "search_trace": trace
    }
```

5.5 OUTPUT

The webapp output:





Task ID	Task	Selected c	Location	Logic	Estimated	Estimated cost
T1	Assess aff	Ground as	Control Ce	OR	0	650
T2	Evacuate l	High-grou	Lowland V	OR	310	3200
T3	Evacuate l	Foot team	Market Di	OR	180	1900
T4	Set up me	Riverside	Riverside	OR	70	1800
T5	Transport	Boat medi	Central De	OR	24	1800
T6	Distribute	Water-tru	Water Pla	OR	500	1500
T7	Open tem	South Sch	North She	OR	670	900
				TOTAL	1754	11750

6. CONCLUSION

6.1 SUMMARIZATION

The ResQPlan Emergency Disaster Response System demonstrates the application of Artificial Intelligence techniques to emergency planning using AND/OR graphs and cost-based decision-making. The system represents emergency locations, tasks, alternative strategies, task dependencies, and estimated costs in a structured manner.

The AND/OR graph represents mandatory activities and alternative actions, while cost evaluation helps identify suitable strategies for completing emergency tasks. The FastAPI-based web application provides an interactive interface for viewing the graph, examining task information, configuring the budget, and displaying the generated response plan.

The project illustrates how graph-based planning can support the organization of emergency operations and help identify cost-effective response strategies under a given budget.

6.2 LIMITATIONS

The current system uses hypothetical locations, estimated costs, and beneficiary values for academic demonstration. Therefore, its results cannot be directly used for real emergency operations without reliable field data and validation.

The planning process depends on the accuracy of the task dependencies, strategy costs, and feasibility information supplied to the system. Real-world factors such as changing weather conditions, road accessibility, resource availability, vehicle capacity, and the urgency of individual rescue operations are not fully represented.

Furthermore, the current planner performs simplified cost evaluation and alternative selection rather than a complete heuristic-guided AO* search with iterative node expansion and cost back-propagation. Consequently, the application should not be considered a guarantee of a globally optimal emergency-response plan.

6.3 FUTURE WORK

The system can be improved by implementing the complete AO* algorithm with heuristic estimates, node expansion, and cost back-propagation to search the AND/OR graph systematically. This would enable more rigorous evaluation of alternative response strategies.

Future development may incorporate real-time weather information, flood maps, road conditions, rescue-team availability, vehicle capacities, and shelter occupancy. Additional constraints can be introduced to account for task urgency, safety requirements, resource limitations, and changing emergency conditions.

The application can also be extended with geographical route visualization, dynamic budget allocation, priority-based scheduling, and integration with external emergency-management systems. These improvements would make ResQPlan more realistic, adaptable, and useful for studying cost-effective disaster-response planning.